

EE 446: TinyML — Lab 5 Questions

Question 1

Raw audio waveforms don't capture frequency patterns well for ML models. Converting to spectrograms/MFCCs creates a 2D time-frequency representation that highlights the unique acoustic features of each keyword, making it much easier for the CNN to distinguish between "yes", "no", silence, and unknown words.

Question 2

The representative dataset is used to calibrate the quantization ranges during INT8 conversion. It shows the converter the real range of activation values so it can accurately map float values to integers without losing too much accuracy.

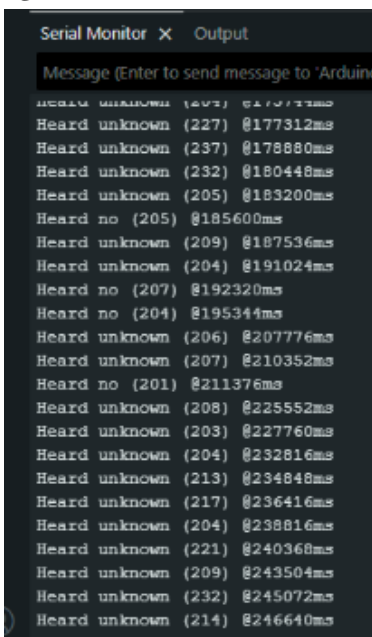
Question 3

`audio_processor.get_data()` loads audio samples from the dataset, applies augmentation (time shifts and background noise), runs feature extraction (MFCC/spectrogram), and returns the processed feature arrays and labels ready for model input or evaluation.

Question 4

`g_model` is a C byte array containing the TFLite model binary, and `g_model_len` is its size in bytes. Since Arduino has no filesystem, the model must be embedded directly in firmware. TFLite Micro uses these to load and run the model from flash memory at runtime.

Question 5



```
Serial Monitor X Output
Message (Enter to send message to 'Arduino')
Heard unknown (227) @0177312ms
Heard unknown (237) @0178880ms
Heard unknown (232) @0180448ms
Heard unknown (205) @0183200ms
Heard no (205) @0185600ms
Heard unknown (209) @0187536ms
Heard unknown (204) @0191024ms
Heard no (207) @0192320ms
Heard no (204) @0195344ms
Heard unknown (206) @0207776ms
Heard unknown (207) @0210352ms
Heard no (201) @0211376ms
Heard unknown (208) @0225552ms
Heard unknown (203) @0227760ms
Heard unknown (204) @0232816ms
Heard unknown (213) @0234848ms
Heard unknown (217) @0236416ms
Heard unknown (204) @0238816ms
Heard unknown (221) @0240368ms
Heard unknown (209) @0243504ms
Heard unknown (232) @0245072ms
Heard unknown (214) @0246640ms
```